

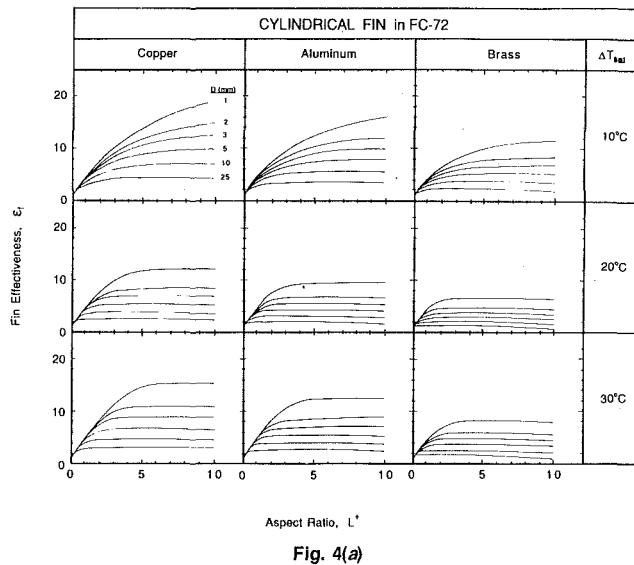


Fig. 4(a)

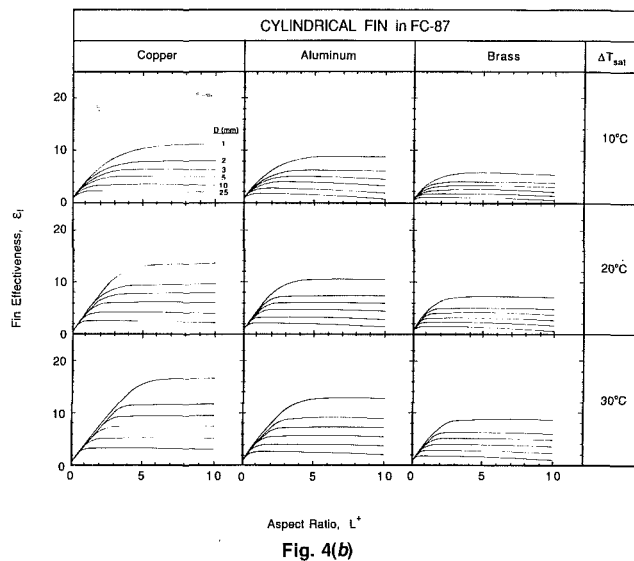


Fig. 4(b)

Fig. 4 Effectiveness plots for heat transfer from a horizontal smooth-surfaced cylindrical fin to (a) saturated FC-72 and (b) saturated FC-87

The numerical results are presented in Figs. 4 and 5 as plots of fin effectiveness, ϵ_f , versus aspect ratio, L^+ , for a given saturated fluid with fixed fin thermal conductivity and base superheat, ΔT_{sat} . The fin effectiveness is defined as the total heat transfer rate from the fin, Q_f , normalized by the heat transfer rate from the reference flat or low-profile microfinned surface, Fig. 3, having a planform area and temperature equal to those of the fin base.

$$\epsilon_f \equiv \frac{Q_f}{q_b A_b} \quad (5)$$

The experimentally determined values of q_b are given in Table 3 for several fluid/surface combinations and four values of base superheat. Fin efficiency, η_f , which represents a measure of the reduced performance due to conduction resistance within the fin is defined as

$$\eta_f \equiv \frac{Q_f}{q_b A_f} \quad (6)$$

where A_f is the total heat transfer area of the fin. Thus, effectiveness and efficiency are directly related by the ratio of total heat transfer area to base area,

$$\epsilon_f = A_f^+ \eta_f \quad (7)$$

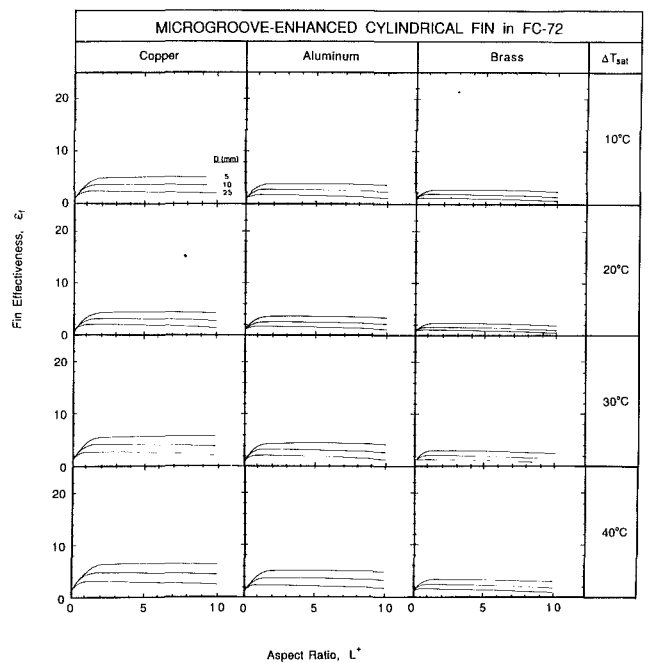


Fig. 5(a)

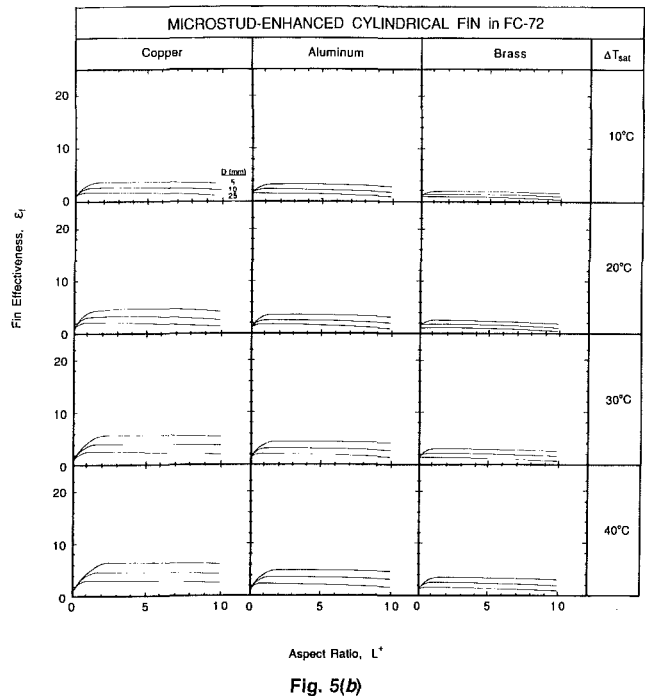


Fig. 5(b)

Fig. 5 Effectiveness plots for heat transfer to saturated FC-72 from a horizontal cylindrical fin with (a) microgroove and (b) microstud enhancement

For a given thermal conductivity and base superheat, Figs. 4 and 5 show effectiveness increases with aspect ratio to a peak value after which it remains constant or decreases. The maximum value of effectiveness is indicative of a trade-off between increased surface area and increased conduction resistance. Increasing fin diameter promotes peak effectiveness values at lower aspect ratios for a related reason. The tip of a large diameter fin comprises a large fraction of total fin heat transfer area and, therefore, convects a large portion of the heat. This moves the peak effectiveness point toward lower aspect ratios compared with small diameter fins. Decreasing the thermal conductivity leads to decreased effectiveness for a fixed ge-